



Maritime &
Coastguard
Agency

Guidance

Annex 3 - Competence Standard for Sea Towage Assist Endorsement

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A3.1 Summary

This standard about:

The competence required to conduct sea towage operations.

The standards for the **Sea Towage Endorsement** are in addition to the requirements of the **General Towage Endorsement**.

This standard is for:

Masters of vessels engaged in sea towage operations.

A3.2 Outcomes of effective performance

Candidates for the issue of the Sea Towage Endorsement will have demonstrated that they can do the following:

1. Ensure that there are clearly defined safe work procedures/standing orders for conducting safe sea towage operations.
2. Inspect the vessel to be towed and establish correctly that it is safe to be towed to the planned destination in the expected sea and weather conditions.
3. Inform the relevant authority promptly where you assess the vessel to be unfit for towing, stating the reasons clearly.
4. Where necessary, ensure that all statutory, regulatory and other requirements for the tow are identified and met, and comply with all navigational, reporting and warning requirements.
5. Assess the circumstances affecting the requirement for a tow and determine correctly any risks involved.
6. Plan the tow, identifying correctly the actions required, taking into account the characteristics of the tow, the expected conditions and any relevant navigational or pilotage restrictions, determining the best method for towing.
7. Establish and maintain clear and appropriate communications between your vessel and the tow.
8. Brief all concerned clearly and correctly on the actions to be taken, ensuring that everyone understands their role and the emphasis upon safety.

9. Prepare all necessary equipment for towing and rigging, checking for damage prior to taking the tow, making good and reporting any faults before use.
10. Develop and action appropriate contingency plans that ensure the safety of navigation, protection of the marine environment and safety of the tug and tow, and persons on board both vessels.
11. React safely and appropriately to loss of critical systems e.g. loss of main power or steering unit whilst engaged in sea towage operations.
12. Identify where the transfer of a crew member to the vessel to be towed would assist the towing operation, and where relevant, undertake such a transfer safely and correctly.
13. Rig, secure and deploy the towing equipment correctly, taking necessary actions to protect both vessels and to minimise the effect of chafe
14. Ensure that the correct audible and visual signals indicate towing.
15. Handle the vessel safely under the prevailing sea conditions with due consideration to the engineering systems.
16. Operate remote controls of propulsion plant, engineering systems and services correctly and safely.
17. Monitor the tow and make any necessary adjustments correctly and safely.
18. Establish communications with those receiving the tow and other relevant authorities and ensure that preparations for receiving the tow on arrival are in place.
19. Disengage the tow correctly and safely.

A3.3 Knowledge and understanding

Underpinning Knowledge Syllabus for Sea Towage Endorsement

Candidates need to know and understand the following:

ST 1 Pre-Towage Inspection

1.1-3 How to carry out an effective assessment of vessel to be towed, to include dimensions, towing points, fixed equipment and cargo.

- 1.4 How to obtain contact details of agents.
- 1.5 Identify Pilotage requirements for departure and arrival ports.
- 1.6 If more than one tug towing, one to be designated as the main towing tug and to be responsible for navigation and ensuring the towing plan is executed.
- 1.7 Up to date weather report for next 24 hours and longer where necessary
- 1.8 Assess the need for additional tugs / guard vessels e.g. When towing long pipelines.
- 1.9 Dangers when operating in close proximity to other vessels, interaction and when using multiple tugs with differing propulsion and bollard pull.
- 1.10 Effects of high windage on a tow and the correlation of windage, expected weather and required bollard pull.
- 1.11 Required statutory certification for the tow
- 1.12 The importance of establishing protocol between lead and subsidiary tugs
- 1.13 Towing vessel is to have sufficient fuel for the voyage
- 1.14 Factors affecting scope of gear to be used during different phases of the tow.

ST 2 Inspection of Tow

- 2.1 Watertight integrity - ensuring vents, watertight doors, hatches, port lights, deadlights, etc are closed
- 2.2 Draft and trim
- 2.3 Sufficient positive stability for the voyage - minimise slack tanks
- 2.4 Engine room inlet and outlet valves – ensuring all are closed
- 2.5 Sea fastenings
- 2.6 For a self-propelled vessel under tow, ensure rudders and shafts secured
- 2.7 Towage arrangements
- 2.8 Navigation lights, daylight shapes and sound signalling appliances
- 2.9 Emergency towing arrangements
- 2.10 Assess means of safe access to the tow at sea.

ST 3 Conduct of the Tow

3.1 Heaving in and paying out towlines and adjustment of engine power

3.2 Catenary of the tow wire – assessing the need to shorten up and reduce power in shallow waters.

3.3 Dangers of large and rapid alterations of course leading to the slack towline catching the seabed - need for sufficient turning circle

3.4 Need for slow and controlled alterations in power applied

3.5 Towing in bad weather, use of tow line length and engine power and heaving to

3.5 Contingency plans for seeking shelter or safe haven.

3.6 Use of appropriate gog arrangement. Safe anchoring arrangements for tug and tow

3.7 Behaviour of towed vessel

3.8 Giving way

3.8 Knowledge and application of Collision Regs:

Crossing traffic lanes

Use of restriction lights and shapes

3.9 Tidal effects

3.10 Monitoring of weather forecasts.

3.11 Checking for chafe and chafe avoidance on passage, Avoidance of use of towing pins to control tow wire

3.12 appropriate reactions in the event of loss of critical equipment (Propulsion, steering).

ST 4 Towing Equipment

4.1 All towing equipment is to be tested and have test certificates held on board

4.2 Industry standard sizing of ropes, wires, shackles and etc for sea towage in relation to the bollard pull of the tug. This is normally at least 3 times the designed bollard pull of the tug with appropriate sized grommets or pennants designed to absorb shock loading of tow lines

- 4.3 Sea towage scope on ropes and on wires
- 4.4 Towing winch operation, streaming and recovering tow
- 4.5 Tow wire sizes and lengths, catenary of the towing wire
- 4.6 Minimisation of chafe at the tug and tow
- 4.7 Emergency towing gear arrangements

ST 5 Fitness for Purpose of Tug, Tow and Crew Competence

- 5.1 Suitable means of access to the tow is to be considered.
- 5.2 Broad white band at bow of tow above waterline for checking from distance.
- 5.3 Rigging and streaming of emergency tow with breakaway fastenings for manned and unmanned tows.
- 5.5 Tow to have a Load Line Certificate, or a Load Line Exemption Certificate from the appropriate Flag State Administration and all conditions to be complied with.
- 5.6 Warranty survey factors:
 - certificates required for tug and tow, towing gear and tug crew
 - conditions to be complied with
 - legal status of warranty survey – for the insurers of the tow and does not take precedence over Flag or Class requirements

ST 6 Emergency Procedures

The tug is to have emergency procedures to enable the crew to effectively deal with the following situations:

- 6.1 Towline failure to include consideration on action to take for:
 - 6.1.1 Recovery of parted line to tug, including section remaining at tow
 - 6.1.2 Clearing tow winch and rigging new tow line
 - 6.1.3 Picking up of emergency tow, connecting to tug's gear
- 6.2 Water ingress into tow.
- 6.3 Failure of sea fastening or other movement of equipment on tow.

ST 7 Importance of Passage Planning when Undertaking a Sea-Tow

7.1 The planning that is required by the tug master, taking into account the characteristics of the tow when considering ports of refuge and the need for necessary third-party tug assistance, disconnection and tying up of the tow at the places of departure and arrival.

ST 8 Importance of Clear and Effective Communication Methods

Tug Master to ensure clear lines of communication and agree with crew necessary protocol during operations with particular reference to crew members and 3rd parties put on board the tow:

- a) During connection phase,
- b) During tow (if applicable)
- c) If connection lost during tow,
- d) During disconnection of the tow.

